SHOU-JEN (LAWRANCE) CHEN

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EDUCATION

Electrical Engineering & Computer Science, University of California, Berkeley (UCB) — **Berkeley, California**

Master of Engineering, Area of Concentration: Robotics and Embedded Software — *Expected Date: Oct 2025 - May 2026*

- GPA: **3.882**/4.0 (Selected Courses: Machine Learning, Embedded Systems, Deep RL, Data Science)

Electrical Engineering, National Taiwan University (NTU) — **Taipei, Taiwan**

Bachelor of Science in Engineering — *Sep 2020 - Jun 2025*

- Dean's List: 2023 Fall (**Top 5%**) & **NTU Admissions Scholarship** for Overseas Chinese Students

Core Skills: Python, Machine Learning, Data Structures & Algorithms, PyTorch, Reinforcement Learning, C++, MuJoCo, ROS

PROFESSIONAL EXPERIENCE

Software Development Engineer — **Seattle, Washington**

Kubernetes Networking, ***Amazon (AWS)*** — *Sep 2026 – Present*

- Software Development Engineer on the Node Runtime team, building networking infrastructure for Kubernetes workloads AWS.

Industry/Lab Researcher — **Taipei, Taiwan**

Industry-University Collaboration, ***TrendForce Corporation*** — *Oct 2024 – Jun 2025*

- Developed automated content generation service integrating LLM APIs with structured prompt templates.
- Built LLM-powered text generation workflow with controllable style parameters and automated quality evaluation loops.
- Implemented style-controlled generation via parameterized prompts and evaluation loops for TrendForce newsletter.

Software Engineer Intern — **Taipei, Taiwan**

Software Engineer Internship, ***Pegatron Corporation*** — *Jun 2023 – Sep 2023*

- Collaborated in a team of three to develop "*Visual Prompt Defect Detection*" for an internal internship project competition, receiving ***1st Place*** among four total groups. Our project focuses on using in-context learning.
- Designed UI using Gradio for project demo & built backend data ingestion pipeline storing image datasets to storage via APIs.
- Completed training on **Git**, **Docker**, **Kubernetes**, **Python**, **Pandas**, **Numpy**, and **MongoDB**.

Software Engineer Webpage Developer Intern — **Taipei, Taiwan**

Software Engineer Internship, ***IADIY Photonics*** — *Jun 2021 – Sep 2021 / Jun 2022 – Aug 2022*

- Designed software enabled auto-driving and basic sensor mapping using Robot Operating System on Linux.
- Assembled robotic devices, including constructing the hardware framework and configuring wires, running on Raspberry Pi.
- Initiated a Kickstarter Campaign in a team of 5 for the product, Genix Lite and designed company website using HTML and CSS.

PROJECTS

Risk-Triggered Latent-Space Corrections for Near-Failure Recovery — **Berkeley, California**

Reinforcement Learning Researcher, ***UC Berkeley*** — *Feb 2026 – May 2026*

- Designed a decoupled correction architecture for Soft Actor-Critic (SAC): a lightweight MLP applies additive latent corrections Δz to the actor's hidden state, gated by a geometric near-failure trigger to only **2–12%** of steps at convergence.
- Applied stop-gradient on the actor encoder to prevent co-adaptation and preserve independent base actor capability.
- Evaluated across 4 Meta-World manipulation tasks; improved mean final success rate by **+57pp** on push-v3 (32.5%→89.5%) and **+27.5pp** on pick-place-v3 (66.0%→93.5%), achieving full seed reliability (4/4) versus 0/4 under concurrent training.

AI-driven Personalized Assisted Device — **Berkeley, California**

Machine Learning Engineer / AI Engineer (Capstone Project), ***UC Berkeley*** — *Aug 2025 – May 2026*

- Built a 3-phase LLM-guided optimization pipeline over a 15-dimensional design space, converging to a personalized assistive device design in as few as 2 iterations and achieving **83% overall convergence rate** across 12 runs and 4 participants.
- Designed task interfaces across 8 simulated grasping environments, enabling evaluation of grip force and tasks success rates.
- Optimized designs reduced required grip force to a mean of 2.5 Nm, with **7 of 8 tasks achievable at or below 1.7 Nm**.

IMU-Controlled Quadruped Robot — **Berkeley, California**

Machine Learning Engineer / Robotics Engineer, ***UC Berkeley*** — *Aug 2025 – Dec 2025*

- Implemented a 4-stage locomotion pipeline (sinusoidal trajectory, planar IK, gait coordination, motor dispatch) supporting **5 gaits** including trot, walk, skid-steer turning, and jump, with stride up to 50 mm and forward velocity of **17 cm/s**.

- Debugged Raspberry Pi 4 to Dynamixel RX-24F UART communication, synchronizing all 8 joints at **100 Hz** with **0.29°** per-tick resolution over a **1 Mbps** daisy-chained bus.
- Built an FSM command dispatcher routing IMU/EMG outputs to joint trajectories; evaluated SVM and DNN classifiers on **8-channel sEMG** at 250 Hz, reaching up to **40%** gesture accuracy with IMU-threshold control deployed for the final demo.

LangGraph ML Research Buddy **Personal Project**

Machine Learning Engineer, ***Personal Project*** *2026*

- Built a stateful six-agent LangGraph pipeline (planner, log analyzer, claim checker, reward-hacking auditor, critic, writer) with conditional edges, bounded retry/reflection loops, and a human-in-the-loop checkpoint for ML experiment review.
- Backed by `SqliteSaver` for durable execution, Pydantic structured outputs per node, a custom `BaseCallbackHandler` tracing LLM calls to per-run artifacts, and a FastAPI/SSE streaming interface; provider-agnostic across Anthropic, OpenAI, and Ollama.

RAG-Forge: Modular RAG Pipeline Framework **Personal Project**

Machine Learning Engineer, ***Personal Project*** *2026*

- Designed and implemented five end-to-end RAG pipelines from naive dense retrieval to graph-based, agentic, and LangChain LCEL architectures, with a provider-agnostic interface supporting Anthropic API, OpenAI API, and Ollama.
- Implemented hybrid retrieval (BM25+dense search, RRF fusion, cross-encoder reranking, HyDE).

Vision Fairness Analysis — CNN vs Vision Transformer **Berkeley, California**

Machine Learning Engineer, ***UC Berkeley*** *Aug 2025 – Dec 2025*

- Designed a fairness study comparing U-Net encoders (ResNet50 vs. FastViT, ImageNet-1k pretrained) on 4× super-resolution and racial classification across FairFace (108k images) and RFW, simulating **1:20 minority-to-majority imbalance** per race.
- Fine-tuned via staged layer unfreezing using perceptual and cross-entropy losses; ResNet50 achieved +**5.79 dB higher PSNR** than FastViT, with both models showing ≤**0.05 dB** per-race variance, confirming super-resolution is race-agnostic.

Smart Parking **Taipei, Taiwan**

Robotics Engineer, ***NTU*** *Aug 2023 – Jan 2023*

- Collaborated with a teammate to develop a smart driving system, featuring real-time notifications and auto-parking.
- Integrated PiRacer Pro and STM32 board to boost system capabilities with analytics and imitation learning.

PUBLICATIONS & CONFERENCE EXPERIENCE

- **Co-First Author**, **COLM 2024**, "*LLM Discussion: Enhancing the Creativity of LLMs via Discussion Framework and Role-Play*"
- **Co Author**, **ICLR 2025**, "*Dynamic-SUPERB Phase-2: An Open Benchmark for Evaluation of Spoken LMs with 180+ Tasks*"
- **Reviewer**, **2024 IEEE Spoken Language Technology Workshop (SLT)**, Reviewed and evaluated research paper.

RESEARCH EXPERIENCE

Robot Learning Laboratory (RLL), NTU **Taipei, Taiwan**

Research Assistant / Undergraduate Researcher | Advisor: ***Prof. Shao-Hua Sun*** *Oct 2023 – Jun 2025*

- Introduced *LLM Discussion* framework that enhances the creativity of large language models via multi-agent discussion and role-playing, surpassing single-agent and existing multi-agent LLM frameworks in creativity tests by 20%.
- Designed an LLM evaluation focused on creativity metrics, achieving a 0.7 correlation with human on 1,400+ responses.
- Develop a decision-making agent that optimizes selection between in-context learning with Retrieval-Augmented Generation, supervised fine-tuning, and reinforcement learning from human feedback in a streamline learning process.
- Investigated test-time compute scaling of algorithmic time complexity (Big-O) and the number of tokens.

Speech Processing and Machine Learning Laboratory (SPMLLab), NTU **Taipei, Taiwan**

Undergraduate Researcher | Advisor: ***Prof. Hung-yi Lee*** *Jul 2023 – Jun 2025*

- Devised and preprocessed a "SoundEffectDetection" dataset/task for Dynamic-SUPERB, aimed at building universal speech models capable of leveraging instruction tuning to perform multiple tasks in a zero-shot fashion.
- Utilize Direct Preference Optimization (DPO) as a method for Reinforcement Learning from Human Feedback (RLHF) to train a text-instruction-guided voice conversion model, aiming to enhance model performance.

Wireless and Mobile Networking Laboratory (WMNLab), NTU **Taipei, Taiwan**

Undergraduate Researcher | Advisor: ***Prof. Hung-Yu Wei*** *Jan. 2023 – Jan. 2024*

- Conducted mobile phone experiments on the MRT using user datagram protocol to evaluate network handover.
- Analyzed data and presented findings on experiments across a 25-km route to professors and graduate students.